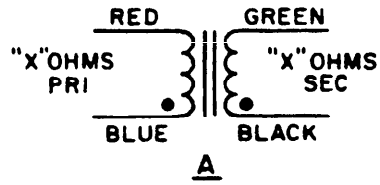
MILITARY SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY, TF5R21ZZ

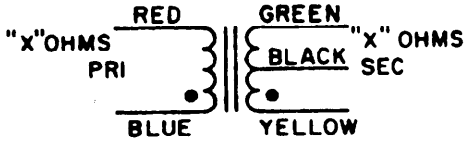
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27.

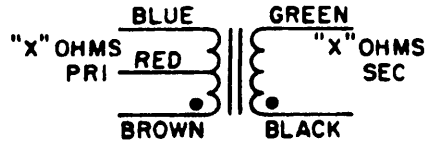
INCHES	MM
.016	.41
.035	.89
.250	6.35
.344	8.74
.469	11.91
1.500	38.10



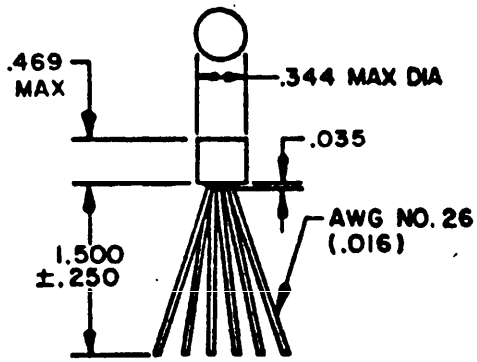
A

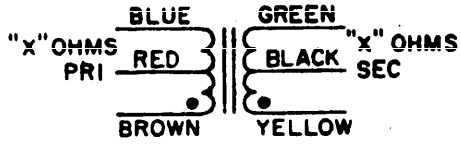


B

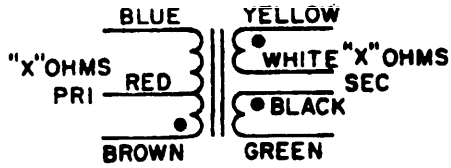


C





D



E

WORKING VOLTAGE (PEAK): 50 V
ALTITUDE: 70,000 FT
(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on the side of the case.
4. Unless otherwise specified, tolerance is ±.005 (.13 mm).
5. Primary and secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS: (When colors in parentheses, i.e., (Brn-Blu), are used they indicate the winding and the extreme terminals of the winding. When the extreme terminals of two windings are used, i.e., (Brn-Blu), the windings are connected in series.)

Electrical ratings: See table I.
Frequency range: 300 Hz to 100 kHz.
Working voltage (peak): 50 volts.

Design and construction:
Dimensions and configuration: See figure 1.
Duty cycle: Continuous.
Case: Epoxy.
Altitude: 70,000 feet, maximum.
Terminals: Insulated wire leads.
Diameter: 0.016 ±0.002 inch.
Length: 1.500 ±0.250 inches.
Weight: 3.0 grams, maximum.
Operating temperature range: -55° to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:
At sea level: 200 volts rms.
At barometric pressure: 100 volts rms.

Vibration (high frequency): MIL-STD-202, method 204, test condition D.

Electrical characteristics: See table II.
No load (centertap voltage unbalance only): With 1 V, 5 kHz across the primary, the voltage unbalance shall be a maximum of 3 percent.
Harmonic distortion: Total harmonic content of output shall be a maximum of 5 percent at the specified power level and primary DC current at 1 kHz (see table I).
Frequency response: At the rated source and load impedance (see table II) and with a 1 milliwatt level output over a frequency range of 300 Hz to 100 kHz, frequency response shall be a maximum of ±3 dB.
Reference frequency = 1 kHz.
Insertion loss: At the specified power level (see table I), the insertion loss shall be a maximum of 3.0 dB.

Marking location: See figure 1.

Part number: M27/144- (dash number from table I).

Supersession data:
MIL-T-27/144 supersedes the following MS sheets which are inactive for design after the date of this specification sheet: MS21416, MS21417, MS21418, MS21420, MS53225, and MS53226.
Superseding part numbers: See table III.

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TABLE I. Electrical ratings. 1/

Dash No.	Circuit diagram (see figures)	Primary impedance (ohms)		Secondary 2/ impedance (ohms)		Primary DC current (mA)	Primary DC resistance +25% (ohms)	Primary power level at 1 kHz (max) (mW)	Secondary DC resistance +25% (ohms)
01	E	80 CT	(Brn-Blu)	32 SPLIT	(Yel-Grn)	12	9.8	600	6
02	C	300 CT		600	(Blk-Grn)	7	19		89
03	E	400 CT		40 SPLIT	(Yel-Grn)	8	46		8
04	E	400 CT		400 SPLIT		8	46		74
05	D	400 CT		4,000 CT		8	45		550
06	A	500	(Blu-Red)	50	(Blk-Grn)	3	60		8
07	C	500 CT	(Brn-Blu)	600	(Blk-Grn)	5.5	31		90
08	D	600 CT		600 CT	(Yel-Grn)	4	47		47
09	C	900 CT		600	(Blk-Grn)	4	52		89
10	A	1,000	(Blu-Red)	50		3	110		8
11	A	1,200		3.2		2	100		.5
12	C	1,500 CT	(Brn-Blu)	600		3	85		89
13	E	2,000 CT		2,000 SPLIT	(Yel-Grn)	3	195	600	125
14	E	2,000 CT		8,000 SPLIT		3	195	125	450
15	B	10,000	(Blu-Red)	500 CT		1	870		50
16	B	10,000		1,200 CT			780		110
17	D	10,000 CT	(Brn-Blu)	1,500 CT			780		125
18	E	10,000 CT		2,000 SPLIT			550		225
19	D	10,000 CT	(Blu-Red)	2,000 CT			1,000		200
20	D	10,000 CT	(Brn-Blu)	10,000 CT			975	125	1,175
21	A	20,000	(Blu-Red)	800	(Blk-Grn)	0.5	825	60	110
22	D	20,000 CT	(Brn-Blu)	800 CT	(Yel-Grn)	0.5	830	60	110

1/ Complete qualification testing to M27/144-20 shall be sufficient to grant qualification approval to M27/144-01 through M27/144-22.

2/ Where windings are listed as SPLIT, one-fourth of the listed impedance is available by paralleling the winding.

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TABLE II. Electrical characteristics.

Dash No.	Frequency response: ±3 dB at 400 Hz to 100 kHz and 1 mW		Resonance, self resonant frequency kHz (min)	Polarity: Additive with terminals (below) connected (see figure 1)
	Z_S (ohms)	Z_L (ohms)		
01	80 CT (Brn-Blu)	32 (Yel-Grn)	750	Blu and Yel, and Wht and Blk
02	300 CT (Brn-Blu)	600 (Blk-Grn)		Blu and Blk
03	400 CT (Brn-Blu)	40 (Yel-Grn)		Blu and Yel, and Wht and Blk
04	400 CT (Brn-Blu)	400 (Yel-Grn)	1,000	Blu and Yel, and Wht and Blk
05	400 CT (Brn-Blu)	4,000 CT (Yel-Grn)	750	Blu and Yel
06	500 (Blu-Red)	50 (Blk-Grn)		Red and Blk
07	500 CT (Brn-Blu)	600 (Blk-Grn)		Blu and Blk
08	600 CT (Brn-Blu)	600 CT (Yel-Grn)		Blu and Yel
09	900 CT (Brn-Blu)	600 (Blk-Grn)		Blu and Yel
10	1,000 (Blu-Red)	50 (Blk-Grn)		Red and Blk
11	1,200 (Blu-Red)	3.2 (Blk-Grn		Red and Blk
12	1,500 CT (Brn-Blu)	600 (Blk-Grn)		Blu and Blk
13	2,000 CT (Brn-Blu)	2,000 (Yel-Grn)		Blu and Yel, and Wht and Blk
14	2,000 CT (Brn-Blu)	8,000 (Yel-Grn)		Blu and Yel, and Wht and Blk
15	10,000 (Blu-Red)	500 (Yel-Grn)		Red and Yel
16	10,000 (Blu-Red)	1,200 (Yel-Grn)		Red and Yel
17	10,000 CT (Brn-Blu)	1,500 (Yel-Grn)		Blu and Yel
18	10,000 CT (Brn-Blu)	2,000 (Yel-Grn)		Blu and Yel, and Wht and Blk
19	10,000 CT (Brn-Blu)	2,000 CT (Yel-Grn)		Blu and Yel
20	10,000 CT (Brn-Blu)	10,000 CT (Yel-Grn)		Blu and Yel
21	20,000 (Blu-Red)	800 (Blk-Grn)	500	Red and Blk
22	20,000 CT (Brn-Blu)	800 (Yel-Grn)	500	Blu and Yel

TABLE III. Supersession data.

Superseding military part number	Superseded MS part number
M27/144-01	MS53225-1
-03	MS21416-1
-04	MS21417-1
-13	MS53226-1
-14	MS21418-1
-18	MS21420-1

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Substitution data (Information only) 1/

Part number	Manufacturer's (FSCM 31669) part number
M27/144-	
01	PICO - A - 1025
02	PICO - A - 1090
03	PICO - A - 1105
04	PICO - A - 1125
05	PICO - A - 1140
06	PICO - A - 1150
07	PICO - A - 1165
08	
09	PICO - A - 1215
10	PICO - A - 1230
11	PICO - A - 1255
12	PICO - A - 1275
13	PICO - A - 1295
14	PICO - A - 1305
15	
16	PICO - A - 1395
17	PICO - A - 1405
18	PICO - A - 1415
19	PICO - A - 1420
20	PICO - A - 1440
21	PICO - A - 1460
22	PICO - A - 1465

1/ Transformers covered by this specification sheet are substitutable for the manufacturer's part numbers. This information in no way implies that the manufacturer's part number is substitutable for the military part number. This information is not intended to be complete.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - SH, OS
Air Force - 11, 17, 19
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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2. REMARKS

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